

Measuring Ownership Changes and Operating Responses in Nursing Homes*

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Abstract

A reported nursing home ownership change need not represent a change in control. We combine transfer dates from Medicare cost reports with monthly records of owners and ownership shares to identify substantive ownership transitions. Among 6,869 U.S. skilled nursing facilities observed from 2017 to 2024, 1,413 meet our transfer definition. Following a transfer, hours per resident day fall for registered nurses and nursing assistants but change little for licensed practical nurses. Total nursing hours change little as the resident census grows, although registered nurse hours fall even in absolute terms. Facilities also become fuller, the Medicare share of resident days rises, the Medicaid share falls, and stays shorten. Measured quality does not broadly worsen. These findings suggest that ownership change alters both nursing skill mix and whom facilities serve; the decline in staffing per resident is one part of a broader change in how facilities operate.

Keywords:

JEL Codes:

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1 Introduction

When a nursing home changes owners, the facility generally remains in the same location, under the same Medicare certification, and subject to the same Medicare and Medicaid payment rules. What changes is control over its operating decisions: whom to admit, how intensively to use nursing labor, how long residents stay, and how to allocate capacity across types of care. These decisions matter because skilled nursing facilities (SNFs) provide two distinct forms of care under very different reimbursement systems. Medicare primarily finances short-term post-acute care, while Medicaid is the principal payer for long-term residential care. Nursing labor is the facility’s largest operating cost, and registered nurses (RNs) are substantially more expensive than other nursing staff. An ownership change therefore leaves many external constraints in place while potentially changing how the facility combines residents, labor, and capacity.

Ownership change is common in this sector ([Prusynski et al., 2023](#); [Oliveira and Blanco, 2024](#))¹ and has drawn growing policy ([Khan, 2024](#); [Murphy and Scanlon, 2026](#)) and public concern ([Americans for Financial Reform Education Fund, 2026](#); [Pifer, 2025](#)), but its effects are not obvious. Facilities change hands for many reasons, including financial distress, succession, and strategic acquisition, and these motives have different implications for staffing, admissions, and quality ([Bowblis, 2011](#); [Lu and Lu, 2022](#)). The direction of the average effect across these motives is therefore theoretically ambiguous. Existing evidence is mixed, and much of it concerns specific forms of ownership change, such as private equity acquisition ([Gupta et al., 2024](#); [Gandhi et al., 2025](#)), chain mergers ([Hirth et al., 2019](#); [Grabowski et al., 2016](#)), or for-profit acquisitions ([Grabowski and Stevenson, 2008](#); [Sutton et al., 2024](#)).

Identifying ownership changes is itself less straightforward than it appears. Administrative datasets identify reported changes in ownership, but a reported change may not represent a substantive transfer of economic control. Changes in legal entities, ownership

¹In our sample roughly one in five SNFs experiences a verified ownership change over seven and a half years.

layers, or reporting arrangements can present as ownership changes while effective control is unchanged. We combine reported ownership-change dates from Medicare cost reports with monthly records of owners and ownership shares to verify substantive transfers of control, and we restrict attention to facilities with at most one verified ownership change. This allows us to study ownership change broadly rather than a particular type of buyer or transaction.

We examine staffing, resident census, payer mix, length of stay, and quality together, and our results reveal a consistent change in how a facility operates after an ownership change. We find that occupancy rises, the census shifts toward Medicare and away from Medicaid, and length of stay shortens. Nursing staff levels, measured in hours per resident day, decline for RNs, certified nursing assistants (CNAs), and in total, and are unchanged for licensed practical nurses (LPNs). Measured in raw hours, only RN hours decline; LPN hours rise, and CNA and total nursing hours are unchanged, so facilities are spreading comparable labor across a larger census. Across quality measures, we see little evidence of broad worsening. Most measures are unchanged, catheter use and urinary tract infections improve, and only weight loss worsens.

This paper makes two contributions. First, it distinguishes reported from substantive ownership changes, which addresses a measurement problem that has constrained this literature. Second, by observing staffing, census, payer mix, and length of stay jointly, it shows that ownership change is better understood as a reorientation of the facility's operating model than as a reduction in nursing labor, and that the standard staffing measure can overstate labor reductions when the census responds. Since the building, certification, and local market are unchanged, these results show how a change in control is associated with a change in how care is produced, holding the local set of providers fixed.

2 Background and Related Literature

An SNF is a healthcare facility that provides skilled nursing care or rehabilitative services and participates in Medicare, Medicaid, or both ([Centers for Medicare & Medicaid Services, 2026b](#)). It provides two forms of care. Short-term post-acute care consists of high-intensity rehabilitation, performed by or under the supervision of professional personnel, to help a patient return home after a hospital stay ([Centers for Medicare & Medicaid Services, 2023](#)). Long-term residential care consists of low-intensity custodial care focused on personal needs and activities of daily living ([Centers for Medicare & Medicaid Services, 2026d](#)). The two are financed by different public payers, both of which set payment rates through legislative or regulatory processes, making SNFs price-takers. Medicare Part A is the primary payer for post-acute care, covering up to 100 days per spell of illness; however, the typical Medicare-covered stay was about 31 days in 2024 ([Medicare Payment Advisory Commission, 2026](#)). Medicaid is the primary payer for long-term residential care, and accounts for the majority of SNF resident days. Rates are set at the state level and are low relative to the cost of care. Some residents initially pay for long-term care themselves, but because of its high cost, most spend down their assets and become eligible for Medicaid early in their stay.

Medicare and Medicaid compensate facilities very differently. The aggregate Medicare margin for freestanding SNFs was 24 percent in 2024, the twenty-fifth consecutive year it exceeded 10 percent, while the non-Medicare margin, driven primarily by Medicaid, was negative 2.3 percent ([Medicare Payment Advisory Commission, 2026](#)). Medicare revenue therefore cross-subsidizes Medicaid losses, and facilities with a higher Medicaid share of patient days face correspondingly weaker finances and staffing capacity ([Bowblis and Brunt, 2025](#)). With prices fixed, a facility's main levers to move profit are how full it is, which payers its residents have, and how quickly it turns residents over.

Nursing labor is a facility's largest operating cost ([Hawk et al., 2022](#)). RNs, including directors of nursing, practice independently and are responsible for clinical assessment, diagnosis, care planning, and delegation; LPNs are responsible for basic and routine nursing

care under the direct supervision of an RN or physician; and CNAs, supervised by licensed nurses, provide most hands-on daily care and are restricted to observation and reporting ([National Council of State Boards of Nursing, 2021](#)). Scope-of-practice rules legally limit substitution between staff types, and compensation follows the same ordering, with an RN hour costing roughly twice a CNA hour ([U.S. Bureau of Labor Statistics, 2023](#)). Staffing is also directly tied to what a facility can deliver. Higher nurse staffing is associated with fewer care-related deficiencies and better resident outcomes ([Harrington et al., 2000](#); [Zhang and Grabowski, 2004](#)), and the association is concentrated in RN staffing ([Lin, 2014](#)).

Federal rules require nursing staff sufficient to meet resident needs, an RN on duty at least eight consecutive hours a day, a licensed nurse on duty around the clock, and a full-time RN director of nursing ([Centers for Medicare & Medicaid Services, 2026e](#)), while state minimum-staffing laws vary in stringency and enforcement.² The federal RN requirement is thin relative to typical RN staffing, much RN work is supervisory and preventive rather than hands-on, and RN hours are the most expensive. Together these features make RN hours a plausible margin of adjustment for a facility seeking to reduce costs, and facilities facing staffing mandates have adjusted by substituting toward less expensive nurse types ([Bowblis and Ghattas, 2017](#)). We do not observe owners' decisions directly, so we treat this as a reason to examine each nurse category separately.

An ownership change unfolds over several months. A buyer and seller agree to a sale, and closing typically waits for state regulatory approval. During this period, the buyer often has some degree of influence over operations, while the seller has an incentive to keep the census and finances stable. A transferred facility may change its name, but nearly all buyers keep the seller's existing CMS Certification Number (CCN) ([Centers for Medicare & Medicaid Services, 2024, 2026a](#)).

Facilities that change ownership are not a random subset of SNFs and differ from those that do not ([Grabowski and Stevenson, 2008](#)). Facilities in financial distress are more likely

²CMS finalized a national minimum staffing rule in 2024 with phase-in scheduled to begin after our sample ends, and that rule was subsequently repealed. The rule never took effect during our sample.

to change ownership (Bowblis, 2011; Lu and Lu, 2022), as are those that are for-profit, chain-affiliated, lower in occupancy, more Medicaid-dependent, and lower-staffed (Prusynski et al., 2023). Evidence on what follows is mixed. Early work finds little change in payer mix, occupancy, or quality after profit-status conversions (Grabowski and Stevenson, 2008), while nonprofit-to-for-profit conversions are followed by cuts to RN staffing and declines in quality (Lu and Lu, 2022). More recent work finds that nurse staffing falls, most sharply for RNs, while the census grows (Ryskina et al., 2024; Prusynski et al., 2025), and that Five-Star staffing and inspection ratings decline after a transfer while the quality-measure rating improves (Ryskina et al., 2024).

We build on this work in two ways. First, we separate staffing per resident day into nursing hours and resident days. A falling staffing ratio can reflect less labor or comparable labor spread across more residents. Second, we examine staffing alongside occupancy, payer mix, and length of stay, which lets us ask whether there is a broader shift in how the facility operates.

3 Measuring Ownership Change

Ownership change is difficult to measure because ownership structures are frequently layered, with an operating entity held by other corporations, trusts, or partnerships. A change in the listed owner of an SNF may not actually correspond to a change in who controls day to day operations of a facility. Legal restructuring, transfers among family members, and changes in reporting conventions can all appear as ownership changes in administrative data without any transfer of operational control.³

Studies on SNF ownership changes generally use one of three administrative sources,

³For example, state Medicaid supplemental payment frameworks in Texas (Texas Health and Human Services Commission, 2026), Utah (Utah Department of Health and Human Services, 2026), and Indiana (Indiana General Assembly, 2026) allow private nursing homes to transfer their licenses to non-state government entities to secure higher reimbursement rates. These are recorded as a change to government ownership in administrative databases, yet the managerial control and operations remain with the original private operators (Office of the Utah State Auditor, 2026; KFF Health News, 2017; KXAN Austin, 2021).

and each, when used alone, can misclassify ownership changes. The Healthcare Cost Report Information System (HCRIS) records whether a facility reported a change in ownership during the reporting period and the effective date of that change. It does not, however, establish transfer of control, so some flagged transfers are changes within the same ownership group. The Provider Enrollment, Chain, and Ownership System (PECOS) records changes of ownership processed through Medicare enrollment. Like HCRIS, it can record reported legal changes rather than changes in control, and it dates them by a tie-in notice that can lag the effective date by several months ([Centers for Medicare & Medicaid Services, 2022](#)).⁴ The Nursing Home Compare (NHC) archive reports monthly owner names, roles, types, and shares, making it possible to observe who owns a facility over time. Its detail, however, can record changes in legal entities that generate false positives.

An approach that other literature takes to sidestep this measurement problem is to study a particular form of ownership change that may be easier to observe. These studies are informative about particular transfers which account for only a portion of ownership change in the sector, and cannot speak to SNF ownership changes as a whole.

We adopt a two-step approach to identify substantive transfers of control. We use HCRIS for the reported change and its date, since its effective date is not subject to the PECOS tie-in lag, and use the NHC Archive to verify that the ownership structure actually changed. Because ownership is layered, we measure changes at the highest ownership level observed for each facility, starting with indirect owners, then direct owners, then partnership-interest owners. We classify a month as an ownership change when more than 50 percent of ownership shares turn over from the previous month. Where ownership percentages are partially or entirely missing, we normalize the reported shares and, where necessary, assign equal weights across the remaining owners. To avoid flagging restructuring within a family, we do not count a change when at least 80 percent of the ownership stake remains within the same family.

⁴PECOS also releases snapshots at points in time that only identify the most recent ownership changes. Therefore, any studies using PECOS (e.g., [Prusynski et al., 2025](#); [Stevenson et al., 2023](#); [Oliveira and Blanco, 2024](#)) can miss ownership changes that may have taken place before the most recently identified change.

Our final sample includes nursing homes that either have no ownership change in either source or have exactly one ownership change in both sources, with the HCRIS and NHC ownership-change dates occurring within six months of one another. For treated facilities, we define the ownership-change month as the month in which HCRIS identifies the change.⁵ Table 1 classifies facilities by the number of ownership changes in each source. Bold cells are facilities that our sample is drawn from. The additional restrictions described in Section 4.1 are applied to these facilities to construct our final sample of 6,869 facilities.

Table 1: Ownership-Change Classification by Source

HCRIS ownership-change status	NHC ownership records		
	0	1	2+
0	6,288	1,874	1,137
1	499	1,759	1,377
2+ changes	15	101	535

Notes: Each cell reports the number of facilities by the number of ownership changes recorded in HCRIS (rows) and identified in the NHC Archive (columns) from January 2017 to June 2024. Counts are before the additional sample restrictions described in Section 4.

4 Data and Methods

4.1 Data and Sample

We combine three administrative data sources. The Payroll-Based Journal (PBJ) reports daily hours worked by staff type and daily resident census derived from the Minimum Data Set (MDS). We aggregate both to the facility-month level. The NHC Archive reports monthly ownership information and chain affiliation, and quarterly CMS quality measures. HCRIS provides occupancy, payer mix, length of stay, certified beds, and reported ownership-change dates. Cost reports are filed once per fiscal year, so we assign each report’s values to every month of its reporting year. When a facility’s outgoing and incoming cost reports both

⁵All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically.

cover the same calendar month, which occurs around an ownership transfer, we average the reported values for that month.

Our sample covers January 2017 through June 2024, classifying ownership changes as described in Section 3. We exclude hospital-based facilities,⁶ facilities in Alaska and Hawaii, facilities with fewer than 15 certified beds, and any facility government-owned at any point during the sample.⁷ We also drop facility-months with implausible staffing: RN and LPN hours both zero, total hours per resident day below 1.5 or above 12, or nurse aide hours per resident day above 5.25 (Centers for Medicare & Medicaid Services, 2026c). Our final monthly sample contains 6,869 facilities and 556,747 facility-months; 1,413 facilities experience a verified ownership change. Our quarterly sample covers 6,857 facilities, 12 less than the monthly sample.

4.2 Outcomes

We measure staffing for three categories using PBJ job codes (Centers for Medicare & Medicaid Services, 2026c): RNs, including RNs with administrative duties and directors of nursing; LPNs, including LPNs with administrative duties; and nurse aides, including certified nursing assistants, aides in training, and medication aides. For facility i in month t , hours per resident day (HPRD) is

$$\text{HPRD}_{it} = \frac{\sum_{d \in t} H_{id}}{\sum_{d \in t} R_{id}}, \quad (1)$$

where H_{id} is the number of hours worked by the staff type in facility i on day d , and R_{id} is the number of resident days on that day. Total HPRD is the sum of the three staff categories.

⁶Hospital-based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than SNF specific incentives (Medicare Payment Advisory Commission, 2026).

⁷We exclude government-owned facilities since transfers to government ownership can reflect supplemental-payment arrangements rather than changes in control (Section 3). Government-owned SNFs also operate in a different financial context (Medicare Payment Advisory Commission, 2026).

We decompose this ratio and estimate effects on both monthly hours and resident days. Quarterly staffing measures for the quality specifications sum hours and resident days over the quarter.

Occupancy is patient days divided by available bed-days; where available bed-days are not reported, we use certified beds times days in the fiscal year. Payer mix is the Medicare and Medicaid shares of patient days, and average length of stay is taken directly from HCRIS.

We use MDS-based quality measures from the NHC Archive. We use eight long-stay measures that fall into two groups. The first reflects care routines that can substitute for staff time. These include: catheter use, antipsychotic medication use, and anti-anxiety or hypnotic medication use. The second group covers resident outcomes: pressure injuries, falls with major injury, weight loss, decline in physical functioning, and urinary tract infections. We also estimate effects on two short-stay measures that come from a different population: new antipsychotic medication and improvement in physical function.⁸ Appendix Table X defines each measure.

Table 2 reports summary statistics for the staffing sample and the quality sample.

⁸Pressure injuries are estimated on 2018–2023 and improvement in function on 2017–2022, the years in which each is reported.

Table 2: Summary Statistics

Variable	Mean	SD	N
Panel A: Staffing sample (facility-month)			
<i>Staffing outcomes</i>			
RN HPRD	0.644	0.372	556,747
LPN HPRD	0.875	0.349	556,747
CNA HPRD	2.265	0.529	556,747
Total HPRD	3.784	0.813	556,747
RN hours (monthly)	1,514	1,127	556,747
LPN hours (monthly)	2,207	1,461	556,747
CNA hours (monthly)	5,621	3,386	556,747
Total hours (monthly)	9,342	5,370	556,747
<i>Facility characteristics</i>			
Non-profit	0.190	0.393	556,747
Chain affiliation (baseline)	0.570	0.495	556,747
Beds	107.5	56.4	556,747
Occupancy rate (%)	77.1	15.4	556,747
% Medicare	13.5	11.4	556,747
% Medicaid	60.1	20.9	556,747
Average length of stay (days)	161.4	159.0	546,692
Acuity quartile 2	0.230	0.421	556,747
Acuity quartile 3	0.233	0.422	556,747
Acuity quartile 4	0.239	0.427	556,747
Panel B: Quality sample (facility-quarter)			
<i>Quality measures</i>			
Catheter use	1.611	2.136	174,616
Anti-psychotic medication use	14.504	9.476	173,209
Anti-anxiety/hypnotic medication use	20.277	10.357	173,808
Pressure injuries	7.923	5.311	123,296
Falls with major injury	3.341	2.831	177,072
Weight loss	6.295	4.886	172,667
Decline in physical functioning	14.811	8.805	165,864
Urinary tract infections	2.510	3.249	176,329
<i>Staffing and facility characteristics</i>			
RN HPRD	0.641	0.366	184,767
LPN HPRD	0.874	0.342	184,767
CNA HPRD	2.260	0.517	184,767
Total HPRD	3.775	0.797	184,767
Non-profit	0.190	0.392	183,676
Chain affiliation (baseline)	0.571	0.495	184,767
Beds	107.8	56.3	184,767
Occupancy rate (%)	77.3	15.2	181,459
% Medicare	13.9	12.7	182,026
% Medicaid	60.2	20.9	164,994
Acuity quartile 2	0.248	0.432	184,767
Acuity quartile 3	0.251	0.434	184,767
Acuity quartile 4	0.258	0.438	184,767

Notes: N is the number of non-missing observations for each variable. HPRD denotes hours per resident day; monthly hours are the HPRD numerator. Chain affiliation is measured at the start of each facility's panel and held fixed. Acuity quartiles are within-state, within-period case-mix quartiles, with quartile 1 omitted. For every quality measure, lower values indicate better measured quality.

Panel A: 6,869 facilities (1,413 with an ownership change), 2017/01–2024/06. Panel B: 6,857 facilities (1,413 with an ownership change), 2017Q1–2024Q2.

Both samples exclude hospital-based facilities, facilities outside the 48 contiguous states, and any facility

4.3 Empirical Strategy

We employ a staggered event-study design that compares changes within facilities before and after an ownership change with contemporaneous changes among never-treated facilities and a treated facility’s own pre-period months. We estimate the following set of event-study regressions,

$$Y_{it} = \sum_{\substack{\tau=-24 \\ \tau \notin \{-4, -3, -2, -1\}}}^{24} \beta_{\tau} \mathbf{1}[T_{it} = \tau] + X_{it}\gamma + \mu_i + \lambda_t + \varepsilon_{it}. \quad (2)$$

where T_{it} is months relative to the ownership change, X_{it} is a vector of controls, μ_i and λ_t are facility and calendar-month fixed effects, and event time runs from -24 to 24 , with the endpoints absorbing all earlier and later periods. In our quality specification, t indexes quarters, λ_t are calendar-quarter fixed effects, and event time runs from -8 to 8 , omitting $\tau = -1$ and $\tau = 0$. We also estimate the following regression,

$$Y_{it} = \delta Post_{it} + X_{it}\gamma + \mu_i + \lambda_t + \varepsilon_{it}. \quad (3)$$

to summarize each outcome with a single post-change coefficient, where $Post_{it}$ equals one for facility-months after the ownership change and δ is the coefficient of interest. Both regressions are estimated on the same samples. Standard errors are two-way clustered by facility and calendar period. Log specifications drop zero values.

Because two-way fixed-effects estimates can be biased by comparisons with already-treated facilities when effects vary over time ([Goodman-Bacon, 2021](#)), we also estimate a stacked design that compares each treated cohort only to never-treated and not-yet-treated facilities.

The recorded date of ownership change is the closing date, but as described in [Section 2](#), the buyer may influence operations while the sale awaits approval. Because staffing can adjust within weeks, the months before closing may already reflect the transfer. We therefore

exclude $\tau = -3, -2, -1$ and use $\tau = -4$ as the reference period in our monthly sample, and exclude $\tau = 0$ and use $\tau = -1$ as the reference period in the quarterly sample. Appendix Tables 7 and 8 report joint Wald tests, which support these choices.

Our choice of controls differs from much of the related literature. Ownership changes plausibly affect facility characteristics that would normally be controlled for, such as occupancy, payer mix, length of stay, acuity, profit status, and chain affiliation. Instead, we treat these characteristics as outcomes, rather than conditioning on them. The only control we use for all specifications is certified beds, which requires state approval to change and rarely does. For quality outcomes, we report specifications that condition on RN, LPN, and CNA HPRD. Because staffing responds to ownership change, these specifications are descriptive rather than estimates of the total effect.

We examine heterogeneity by baseline chain affiliation at the start of our sample.⁹ We use the baseline because chain status frequently changes around the ownership transfer, so the contemporaneous measure is partly a result of treatment. We estimate Equation 3 separately for chain-affiliated and independent facilities. Facilities with no chain classification are excluded from these comparisons.

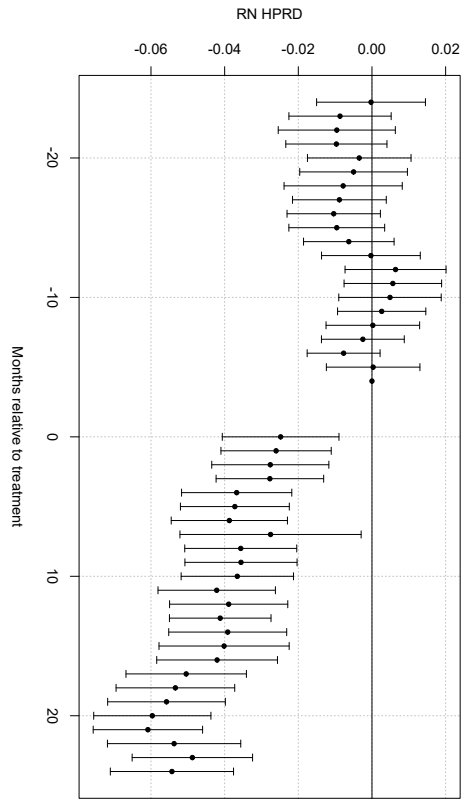
5 Results

5.1 Staffing and Resident Census

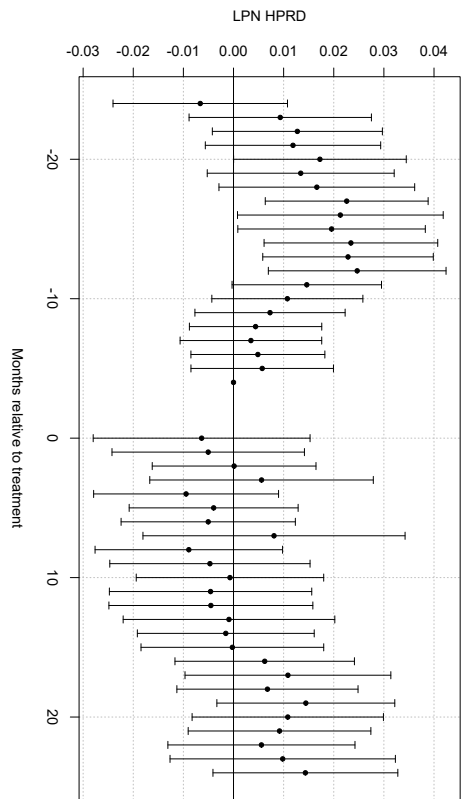
Figure 1 presents event study estimates for nursing HPRD. Before ownership change, the coefficients are small and statistically indistinguishable from zero relative to $\tau = -4$, except for LPNs. After the change, RN, CNA, and total HPRD fall and stay below their pre-transfer levels. LPN HPRD shows little change. The decline appears in the first few months and does not revert, suggesting a lasting change in how facilities staff, rather than just a short disruption around the event.

⁹If unavailable, we define it as the facility's first observation.

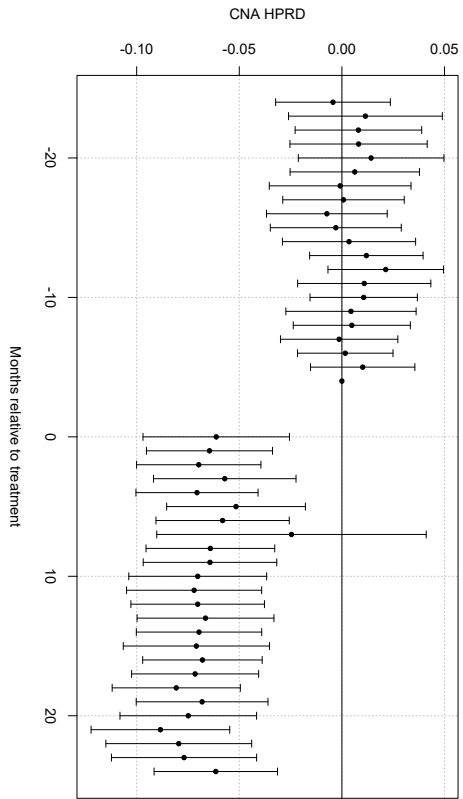
(a) Registered Nurses (RN)



(b) Licensed Practical Nurses (LPN)



(c) Certified Nursing Assistants (CNA)



(d) Total Nursing Hours

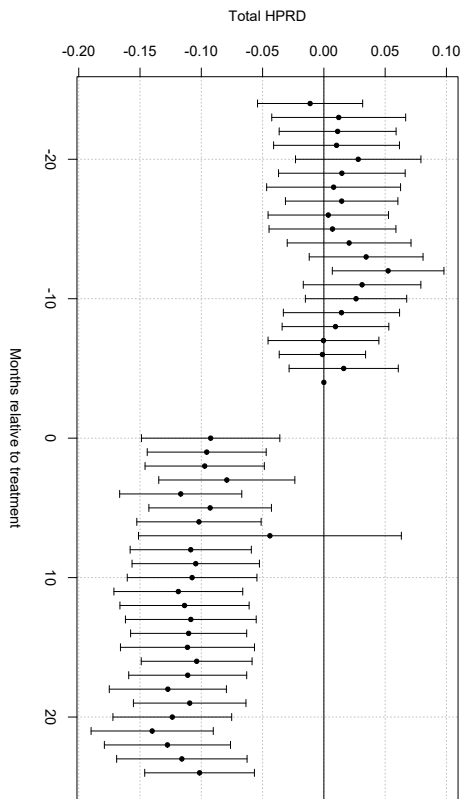


Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per resident day (HPRD). Points show coefficient estimates relative to the reference period $\tau = -4$; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

Table 3 summarizes these patterns with a single coefficient. HPRD falls by 0.042 for RNs, 0.064 for CNAs, and 0.106 in total, or 6.6, 2.8, and 2.8 percent of their sample means. LPN HPRD shows little change, 0.001 and not significant.

The remaining rows of this table separate HPRD into its numerator and denominator, hours and resident census. Resident days rise by 89 per month, or roughly 3.5 percent. Total nursing hours (+82, SE 52) and CNA hours (+52, SE 37) are not statistically distinguishable from zero. LPN hours rise by 95 per month (4.3 percent). Most of the decline in staffing per resident therefore reflects a similar amount of nursing labor spread across more residents. The exception is RN hours, which fall by 64 per month (4.3 percent), even as census grows. RNs are the only staff type whose absolute hours decline, which is why their HPRD reduction is larger than the other staff types.

The staffing response differs by baseline chain affiliation. Total HPRD falls by 0.149 among independent facilities and by 0.079 among chain-affiliated facilities, and the gap comes mainly from CNA staffing (-0.092 versus -0.048). The decline in RN HPRD, by contrast, is similar in the two groups (-0.041 and -0.040), and the difference is not statistically significant (Appendix Table 9).

Table 3: Effect of Ownership Change on Nursing Staffing

Outcome	RN	LPN	CNA	Total
Panel A: Hours per resident day (HPRD)				
HPRD	−0.0423*** (0.0058)	0.0008 (0.0063)	−0.0641*** (0.0100)	−0.1055*** (0.0141)
Log(HPRD)	−0.0722*** (0.0097)	0.0089 (0.0094)	−0.0286*** (0.0046)	−0.0285*** (0.0038)
Panel B: Hours and resident days				
Hours	−64.3765*** (15.2615)	94.6915*** (18.0220)	51.6313 (36.5911)	81.9463 (52.0420)
Resident days		89.3846*** (12.1280)		
Panel C: Chain-affiliated at baseline (N = 315,111 facility-months)				
HPRD	−0.0410*** (0.0072)	0.0108 (0.0077)	−0.0484*** (0.0114)	−0.0786*** (0.0164)
Log(HPRD)	−0.0698*** (0.0116)	0.0195* (0.0116)	−0.0228*** (0.0053)	−0.0226*** (0.0044)
Panel D: Independent at baseline (N = 237,902 facility-months)				
HPRD	−0.0401*** (0.0095)	−0.0161 (0.0113)	−0.0924*** (0.0187)	−0.1487*** (0.0263)
Log(HPRD)	−0.0687*** (0.0171)	−0.0112 (0.0164)	−0.0388*** (0.0087)	−0.0377*** (0.0072)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period.

Panel A reports staffing HPRD. Panel B decomposes each staffing measure into its numerator (raw monthly hours, the total quantity of nursing labor purchased) and denominator (resident days, i.e. facility census).

Panels C and D re-estimate Panel A separately by baseline chain affiliation. Chain status is each facility's classification at its first observation in the panel. Facilities with no available chain classification are excluded from both split panels.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 The Operating Model

While there is a rise in resident days, that is one part of a broader change in how facilities operate after undergoing an ownership change. Table 4 reports estimates for the operating model outcomes. Occupancy rises by 2.7 percentage points. The resident mix shifts towards Medicare, as its share of patient days increases by 1.6 percentage points (roughly 12 percent of its mean). The Medicaid share falls by 1.2 percentage points, although only significant at

the 10 percent level. Average length of stay falls by 13 days (about 8 percent). Acuity????

These results are consistent with the payment environment described in Section 2. Since SNFs are price-takers and those prices are set administratively, the main ways a facility can change its revenues are how full it is, who pays for its residents, and how quickly it turns residents over. After an ownership change, all of these outcomes are consistent with an increase in Medicare resident mix. The facility is fuller, has more post-acute residents (Include?), and shorter stays. Since these outcomes come from annual cost reports, their timing is less precise than the monthly staffing estimates.

Most of the change in the operating model occurs at facilities that were independent at baseline. Occupancy rises by 3.9 percentage points among independent facilities and by 2.0 among chains. The Medicaid share falls by 3.6 percentage points among independent facilities but does not change among chains, and length of stay falls by 22.8 days among independent facilities, compared with a smaller 8.3 among chains. The Medicare share, however, rises by a similar amount in both groups (1.5 and 1.8).

Table 4: Effect of Ownership Change on Business-Model Outcomes

Outcome	Pooled	Chain	Non-chain
Occupancy rate	2.7053*** (0.3506)	2.0236*** (0.4181)	3.9347*** (0.6030)
Medicare share	1.5915*** (0.2160)	1.7716*** (0.2306)	1.4914*** (0.4204)
Medicaid share	-1.1789* (0.5947)	0.1308 (0.7349)	-3.6001*** (0.9633)
Average length of stay	-13.0413*** (4.4632)	-8.2674 (5.1983)	-22.7539** (9.0237)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period.

Occupancy rate is residents as a share of available bed-days. Payer shares are shares of patient days.

The chain and non-chain columns re-estimate the pooled specification separately by baseline chain affiliation, defined as each facility's classification at its first observation in the panel. Pooled sample: N = 553,013 facility-months. Chain: N = 315,111. Non-chain: N = 237,902.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.3 Quality

Table 5 reports coefficient estimates for the quality measures. Appendix Figures 2 and 3 show the corresponding event study estimates.

Of the measures that reflect care routines that can substitute for staff time, catheter use falls by 0.17 percentage points (10 percent of its mean). Antipsychotic (-0.33 , SE 0.23) and anti-anxiety or hypnotic medication use (-0.21 , SE 0.20) also fall, but not significantly. Of the resident outcome measures, urinary tract infections fall by 0.33 percentage points (about 13 percent of its mean). Pressure injuries, falls with major injury, and decline in physical functioning show no significant difference. Weight loss is the only measure that worsens, rising by 0.21 percentage points. Neither short-stay measure changes significantly. Adding RN, LPN, and CNA HPRD as controls has little effect on the estimates. Because staffing is itself affected by ownership change, these specifications are descriptive rather than estimates of the total effect.

Catheter use falls by 0.33 percentage points among independent facilities. Weight loss rises by 0.58 among independents and is flat among chains. UTIs fall in both groups (-0.26 and -0.45). Antipsychotic use falls by 0.68 among independents and functional decline rises by 0.51 among chains.

Overall, measured quality does not broadly worsen after an ownership change. However, these measures are imperfect, and we are not able to observe hospitalization, mortality, or resident experience.

Table 5: Effect of Ownership Change on Quality Measures

Outcome	(1)	(2)	(3)	(4)
Panel A: Long-stay labor-saving mechanism measures				
Catheter use	−0.1665*** (0.0483)	−0.1713*** (0.0484)	−0.0710 (0.0555)	−0.3346*** (0.0857)
Anti-psychotic medication use	−0.3261 (0.2255)	−0.3512 (0.2267)	−0.1633 (0.2688)	−0.6774* (0.3878)
Anti-anxiety/hypnotic medication use	−0.2055 (0.2016)	−0.1748 (0.2009)	−0.3326 (0.2493)	0.1067 (0.3475)
Panel B: Long-stay resident outcome measures				
Pressure injuries	0.2628 (0.1781)	0.2726 (0.1815)	0.3512 (0.2138)	0.0599 (0.2508)
Falls with major injury	0.0113 (0.0634)	0.0135 (0.0642)	0.0509 (0.0781)	−0.0748 (0.1149)
Weight loss	0.2050* (0.1037)	0.2160** (0.1035)	0.0070 (0.1285)	0.5836*** (0.1879)
Decline in physical functioning	0.3329 (0.2716)	0.3546 (0.2812)	0.5050* (0.2951)	0.0399 (0.4238)
Urinary tract infections	−0.3299*** (0.0756)	−0.3149*** (0.0752)	−0.2616*** (0.0893)	−0.4464*** (0.1263)
Panel C: Short-stay measures				
New antipsychotic medication	−0.0208 (0.0431)	−0.0239 (0.0430)	−0.0480 (0.0519)	0.0396 (0.0954)
Improved function	−0.0533 (0.4896)	−0.0609 (0.4913)	−0.2711 (0.5658)	0.1539 (1.0014)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Column (1) is the baseline specification. Column (2) adds RN, LPN, and nurse aide hours per resident day as controls. Because staffing is itself affected by ownership change, column (2) is not a preferred estimate of the total effect; it is reported to show whether the quality response is attenuated after conditioning on measured staffing inputs. Columns (3) and (4) re-estimate column (1) separately for chain-affiliated and independent facilities respectively, defined by each facility's classification at its first observation in the panel.

Long-stay measures (Panels A-B) and short-stay measures (Panel C) are constructed from different resident populations and are not directly comparable to one another. For every measure, lower values indicate better measured quality.

Pressure injuries is estimated on 2018–2023 only; improved function is estimated on 2017–2022 only, the years in which each is reported. Vaccination measures are excluded. The transition quarter ($\tau = 0$) is excluded.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4 Robustness

We assess our results in several ways, and Appendix 8 describes each robustness check in detail. A stacked design that compares each treated cohort only to never-treated and not-yet-treated facilities gives similar estimates and dynamics. The estimates change very little when we change the width of our donut window. They also hold when we exclude facility months that follow gaps in PBJ reporting. In all cases our estimates are similar.

6 Discussion

This paper studies ownership changes in nursing homes as moments of organizational disruption in care production. Our central finding is that ownership transfer is business-model reorientation rather than simple cost containment. Facilities emerge from the transition fuller, shifted toward Medicare, with shorter stays and higher measured acuity. They employ slightly more total nursing labor in absolute terms but spread it across a larger resident census, so staffing per resident falls. Within that larger workforce, composition shifts away from registered nurses, whose hours decline outright while other categories grow. Measured quality does not broadly deteriorate. Taken together, these patterns indicate that ownership changes do more than alter the legal identity of the owner: they change what the facility does and who it serves.

6.1 Reorientation Under Administered Prices

A nursing home cannot compete on price. Medicare and Medicaid per-diems are set administratively, so an owner seeking to improve financial performance must operate on volume, payer composition, and throughput. That is what we observe. Occupancy rises by 2.68 percentage points. The Medicare share of patient days rises by 1.61 percentage points while the Medicaid share falls by 1.17 points. Average length of stay falls by more than thirteen days, and measured acuity rises.

These are not four separate findings but one strategy observed from several angles. Filling empty beds, admitting more post-acute Medicare residents, and turning residents over more quickly are responses that are complementary to a fixed price. A facility can only fill beds that are empty.

This reorientation is concentrated among facilities that were independent at baseline. Occupancy rises by 3.87 percentage points among independents against 2.03 among chains, and both the shift away from Medicaid and the reduction in length of stay are detectable only among independents. Admitting more Medicare residents, by contrast, is common to both. One reading is that independent operators face tighter financial constraints, which makes payer composition a more immediate margin. Another is that they simply had more scope for reorganization to begin with, not having been run under the standardized admissions and throughput practices a chain would already have imposed.

6.2 Interpreting the Staffing Response

The staffing results are frequently summarized in this literature as reductions in nursing labor; however, following that interpretation is incomplete here. Per resident day staffing falls for every category except LPNs. Measured in raw hours, only registered nurse hours decline. LPN and CNA hours both rise, and total nursing hours are statistically indistinguishable from unchanged or marginally higher.

The two sets of estimates are reconciled by the census. Because hours per resident day is hours divided by resident days, the difference between the log-hours and log-HPRD coefficients gives the implied change in the denominator. Estimating separately for each staff category returns census growth of four percent in every case, matching the occupancy increase we show above. Per-resident staffing falls because facilities add residents faster than they add labor, not because they purchase less labor.

Registered nursing does not follow this pattern, and it does so in the way we predict in Section ???. RN hours fall in absolute terms even as the facility grows. The features we

have previously discussed could explain this. The federal floor requiring an RN on duty eight consecutive hours per day is thin relative to typical staffing, so RN hours contain slack that regulation does not protect; RN work is disproportionately assessment, care planning, surveillance, and supervision; and RN hours are the most expensive, so each hour removed yields the largest saving.

Whether these magnitudes are economically meaningful depends on the baseline. Mean staffing in our sample is 0.432 RN hours per resident day, 0.802 LPN hours, 2.118 CNA hours, and 3.352 hours in total. The estimated reductions of 0.037, 0.053, and 0.094 hours per resident day for RNs, CNAs, and total staffing correspond to roughly 8.5 percent, 2.5 percent, and 2.8 percent of baseline. Expressed in time, the total reduction is approximately five and a half fewer nursing minutes per resident-day. These are modest changes and should not on their own be read as evidence of substantial welfare loss. But staffing is a central input into care production, and reductions of this size may matter for facilities operating close to a staffing constraint or for residents requiring more intensive assistance.

Organizational structure shapes the size of this adjustment. Independent facilities reduce per-resident staffing by roughly three times as much as chain-affiliated facilities, and the gap is concentrated in nurse aide staffing. Chains may draw on internal labor markets, standardized staffing practices, and administrative support that allow them to absorb a transition with less disruption, while independent operators face tighter managerial and financial constraints that make staffing a more immediate margin. Registered nursing is the exception to this pattern as well: the level estimates are nearly identical across the two groups, so the margin we identify as most available to a cost-conscious acquirer appears to be used by both types of acquirer alike.

6.3 Interpreting the Quality Response

If ownership change were primarily quality-reducing cost containment, staffing-intensive resident outcomes should worsen. Broadly, they do not. Pressure injuries, falls with major

injury, and decline in physical functioning are not statistically distinguishable from zero. Weight loss shows a marginally significant increase and is the one measure with worsening. Urinary tract infections decline, as does catheter use. The medication measures decline in point estimate but are not estimated precisely enough to support a claim.

The census does not simply grow; it changes. Shorter stays, a larger Medicare share, and higher recorded acuity describe a shift toward post-acute rehabilitative care, and post-acute residents differ from long-stay residents in the care they require. Several of our quality measures are constructed from long-stay assessments, so a facility that reorients toward short-stay care may show stable long-stay quality simply because the long-stay population it retains is smaller or differently composed. On this front, stable quality reflects a change in what is being produced rather than an improvement in how efficiently it is produced.

We cannot separate these possibilities with the measures available: whether quality genuinely held up as the facility grew, or whether the long-stay population being measured changed alongside the census. What the results do establish is that the simplest account, in which new owners cut labor and residents suffer measurable harm, is not what the data show.

The chain split does not reveal a trade-off between the scale of reorganization and measured quality. Independent facilities, which reorganize the most on every other margin, also show the larger measured quality improvements. Each group has exactly one measure that worsens, and they are different measures: weight loss among independents, decline in physical functioning among chains. That the one deterioration on the chain side appears in a measure of functional decline, and is not accompanied by the offsetting improvements observed among independents, sits awkwardly with a straightforward reading in which chains manage transitions more smoothly. We do not read either result as strong evidence on its own. Both are single measures and one is only marginally significant; Appendix Table 9 reports formal tests of whether the chain and non-chain coefficients differ.

6.4 Limitations

Several limitations can shape the interpretation of these results.

First, the timing of ownership change may be measured with error despite our two-source verification. The recorded date is the closing date, and contract negotiation, transition planning, and operational change can precede it. Our preferred staffing specification excludes the three months before closing and our preferred quality specification excludes the transition quarter, but residual timing error may remain.

Second, ownership changes are not randomly assigned. The design addresses time-invariant facility differences and common shocks, but ownership transfer may coincide with unobserved facility-level shocks such as financial distress, management turnover, or local labor market conditions that independently affect staffing and quality.

Third, the business-model outcomes are drawn from annual cost reports and expanded across the months of each fiscal year. They therefore change at most once per fiscal year and are constant within it by construction. This limits the time resolution of any event study estimated on these outcomes and induces serial correlation within facility-year. The direction and magnitude of the average post-change estimates are informative, but the dynamic path of occupancy, payer mix, and length of stay around the transfer is not identified at monthly frequency.

Fourth, the staffing and quality measures are imperfect. PBJ data may contain reporting gaps or measurement error, although the robustness checks indicate that reporting gaps do not drive the staffing results. CMS quality measures capture selected process and resident outcomes but do not measure resident welfare, resident experience, deficiencies, hospitalizations, or mortality.

Finally, these results should not be read as establishing that ownership change improves welfare. The evidence shows that facilities serve more residents with approximately constant labor while measured quality does not broadly worsen. That pattern is consistent with efficiency-enhancing reorganization, but it is also consistent with a change in the composition

of care that our measures do not fully capture, and unmeasured dimensions of resident welfare may behave differently.

7 Conclusion

Ownership changes are common in U.S. nursing homes, yet their consequences for care production remain poorly understood. This paper uses CMS administrative panel data from 2017 through 2024 to identify transfers of substantive economic control, verified across two independent sources, and estimates how facilities change afterwards.

Ownership transfer is followed by a reorientation of the operating model. Occupancy rises, the resident census shifts toward Medicare and away from Medicaid, length of stay shortens, and measured acuity increases. Facilities respond by employing slightly more total nursing labor while serving a census that grows faster than that labor, so staffing per resident falls. The composition of that labor shifts: registered nurse hours decline in absolute terms while licensed practical nurse and nurse aide hours rise. Measured quality does not broadly deteriorate, and two measures, catheter use and urinary tract infections, improve.

Facilities produce care for more residents with approximately constant total nursing labor and without broad measured deterioration in quality. That is consistent with the transfer disrupting established routines and reallocating labor across tasks. It is also consistent with a shift in the kind of care being produced, from long-stay custodial toward short-stay post-acute, that our measures only partly capture. Distinguishing between these is the natural next step for this literature.

The contribution is to treat ownership change as a shock to the internal organization of production rather than to market structure. A facility that changes hands remains in the same location, under the same certification, serving the same local market. What changes is who decides how many residents to admit, from which payer, for how long, and with what mix of nursing labor. In a sector where prices are administratively fixed and labor is the

dominant cost, those decisions are where ownership makes itself felt. Future work should examine which types of transfers generate efficiency-enhancing reorganization and which produce quality-reducing cost containment, and whether the reorientation documented here changes outcomes this paper does not observe, including hospitalization, resident experience, and mortality.

References

- Americans for Financial Reform Education Fund (2026). 2026 polling: Voters deeply concerned about private equity’s impacts and influence. Americans for Financial Reform Education Fund. Accessed: 2026-09-21.
- Bowblis, J. R. (2011). Ownership conversion and closure in the nursing home industry. *Health Economics*, 20(6):631–644.
- Bowblis, J. R. and Brunt, C. S. (2025). Nursing home staffing expenditures and levels are impaired by high medicaid payer-mix. *Journal of the American Medical Directors Association*, 26(8):105723.
- Bowblis, J. R. and Ghattas, A. (2017). The impact of minimum quality standard regulations on nursing home staffing, quality, and exit decisions. *Review of Industrial Organization*, 50:43–68.
- Centers for Medicare & Medicaid Services (2022). *Medicare Fee-For-Service Provider Enrollment - Skilled Nursing Facilities (SNF) Change of Ownership: Data Guidance*. Center for Program Integrity (CPI), Provider Enrollment Oversight Group, ver. 1.0 edition.
- Centers for Medicare & Medicaid Services (2023). *Medicare Benefit Policy Manual, Chapter 8: Coverage of Extended Care (SNF) Services Under Hospital Insurance*. U.S. Department of Health and Human Services, rev. 12283 edition. Section 30.2: Skilled Nursing and Skilled Rehabilitation Services.
- Centers for Medicare & Medicaid Services (2024). *State Operations Manual, Appendix PP - Guidance to Surveyors for Long Term Care Facilities*. U.S. Department of Health and Human Services, rev. 225 edition.
- Centers for Medicare & Medicaid Services (2026a). Change of Ownership, 42 CFR § 489.18. Electronic Code of Federal Regulations (eCFR).
- Centers for Medicare & Medicaid Services (2026b). Definitions, 42 CFR § 476.1. Electronic Code of Federal Regulations (eCFR). Current as of September 18, 2026.
- Centers for Medicare & Medicaid Services (2026c). *Design for Care Compare Nursing Home Five-Star Quality Rating System: Technical Users’ Guide*. U.S. Department of Health and Human Services. Accessed: 2026-09-28.
- Centers for Medicare & Medicaid Services (2026d). *Medicare Benefit Policy Manual, Chapter 16: General Exclusions From Coverage*. U.S. Department of Health and Human Services, rev. 13593 edition. Section 110: Custodial Care.
- Centers for Medicare & Medicaid Services (2026e). Nursing Services, 42 CFR § 483.35. Electronic Code of Federal Regulations (eCFR).

- Gandhi, A., Song, Y., and Upadrashta, P. (2025). Private Equity, consumers, and competition: Evidence from the Nursing Home industry. Working Paper 34306, National Bureau of Economic Research.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Grabowski, D. C., Hirth, R. A., Intrator, O., Li, Y., Richardson, J., Stevenson, D. G., Zheng, Q., and Banaszak-Holl, J. (2016). Low-quality Nursing Homes were more likely than other Nursing Homes to be bought or sold by chains in 1993–2010. *Health Affairs (Millwood)*, 35(5):907–914. PMID: 27140998.
- Grabowski, D. C. and Stevenson, D. G. (2008). Ownership conversions and nursing home performance. *Health Services Research*, 43(4):1184–1203.
- Gupta, A., Howell, S. T., Yannelis, C., and Gupta, A. (2024). Owner incentives and performance in healthcare: Private equity investment in nursing homes. *The Review of Financial Studies*, 37(4):1029–1077.
- Harrington, C., Zimmerman, D., Karon, S. L., Robinson, J., and Beutel, P. (2000). Nursing home staffing and its relationship to deficiencies. *The Journals of Gerontology: Series B*, 55(5):S278–S287.
- Hawk, T., White, E. M., Bishnoi, C., Schwartz, L. B., Baier, R. R., and Gifford, D. R. (2022). Facility characteristics and costs associated with meeting proposed minimum staffing levels in skilled nursing facilities. *Journal of the American Geriatrics Society*, 70(4):1198–1207.
- Hirth, R. A., Zheng, Q., Grabowski, D. C., Stevenson, D. G., Intrator, O., and Banaszak-Holl, J. (2019). The effects of chains on the measurement of competition in the Nursing Home Industry. *Medical Care Research and Review*, 76(3):315–336. PMID: 29148340; PMCID: PMC13159488.
- Indiana General Assembly (2026). Indiana Code § 12-15-14: Medicaid Payment for Nursing Facility Services. Indiana Official State Code.
- KFF Health News (2017). Chasing millions in Medicaid dollars, hospitals buy up nursing homes. KFF Health News. Accessed: 2026-09-27.
- Khan, L. M. (2024). Remarks by chair lina m. khan as prepared for delivery: Private capital, public impact workshop on private equity in healthcare. Federal Trade Commission.
- KXAN Austin (2021). Obscure program sends big money to Texas nursing homes amid pandemic. Is it protecting residents? KXAN Austin. Accessed: 2026-09-27.
- Lin, H. (2014). Revisiting the relationship between nurse staffing and quality of care in nursing homes: An instrumental variables approach. *Journal of Health Economics*, 37:13–24.
- Lu, L. X. and Lu, S. F. (2022). Does nonprofit ownership matter for firm performance? financial distress and ownership conversion of nursing homes. *Management Science*, 68(7):5127–5145.
- Medicare Payment Advisory Commission (2026). Report to the Congress: Medicare payment policy. Technical report, Medicare Payment Advisory Commission (MedPAC), Washington, DC.
- Murphy, C. and Scanlon, M. G. (2026). Murphy, scanlon introduce bicameral bill to ban private equity from owning nurses and hospitals. Press release.
- National Council of State Boards of Nursing (2021). The NCSBN model act and rules. Accessed: 2026-09-22.

- Office of the Utah State Auditor (2026). Medicaid upper payment limit supplement for skilled nursing facilities. Management Letter 25-06, Office of the Utah State Auditor.
- Oliveira, I. and Blanco, M. (2024). Changes in ownership among skilled nursing facilities from 2016 to 2021: Variations by size, occupancy rate, penalty amount, and type of ownership. Research brief, Office of the Assistant Secretary for Planning and Evaluation, U.S. Department of Health and Human Services, Washington, DC.
- Pifer, R. (2025). Private Equity poses ‘new and unique risks’ to Healthcare, federal report warns. Healthcare Dive. Accessed: 2026-09-21.
- Prusynski, R. A., Amaravadi, H., Frogner, B. K., and Mroz, T. M. (2025). Impacts of skilled nursing facility change of ownership on staffing: A staggered difference-in-difference analysis. *Journal of the American Medical Directors Association*, 26(5):105530.
- Prusynski, R. A., Humbert, A., and Mroz, T. M. (2023). Skilled nursing facility changes in ownership and short-stay medicare patient outcomes. *JAMA Network Open*, 6(9):e2334551–e2334551.
- Ryskina, K. L., Tu, E., Liang, J., Kim, S., and Werner, R. M. (2024). Nursing home compare star ratings before versus after a change in nursing home ownership. *Journal of the American Geriatrics Society*, 72(10):3078–3088.
- Stevenson, D., Peterson, H., Skinner, R., et al. (2023). Trends in ownership structures of u.s. nursing homes and the relationship with facility traits and quality of care (2013–2022). Research brief, Office of the Assistant Secretary for Planning and Evaluation, U.S. Department of Health and Human Services, Washington, DC.
- Sutton, N., Ma, N., Yang, J. S., and Lin, J. (2024). Quality effects of home acquisitions in residential aged care. *Australasian Journal on Ageing*, 43(1):158–166.
- Texas Health and Human Services Commission (2026). 1 Texas Administrative Code § 353.1302: Quality Incentive Payment Program for Nursing Facilities. Texas Administrative Code. Available at the Texas Secretary of State Office.
- U.S. Bureau of Labor Statistics (2023). Occupational employment and wage statistics, may 2022. Occupational Employment and Wage Statistics program.
- Utah Department of Health and Human Services (2026). Utah Admin. Code R414-505-5: Requirements to Participate in the NF NSGO UPL Program. Utah Administrative Code. Accessed via Cornell Legal Information Institute.
- Zhang, X. and Grabowski, D. C. (2004). Nursing home staffing and quality under the nursing home reform act. *The Gerontologist*, 44(1):13–23.

Tables

Table 6: Examples of Discordant Ownership Records Across HCRIS and NHC

Example	Classification	Period	NHC owner names and shares
<i>Panel A: HCRIS-reported ownership changes not verified as substantive transfers in NHC records</i>			
A	HCRIS only	$t = 1$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (50%); The Gail Miller GST (50%)
A	HCRIS only	$t = 2$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (20%); The Gail Miller GST (20%); Karen Miller-related trust entities (60%)
B	HCRIS only	$t = 1$	Vetter Eldora (50%); Vetter Jack (50%)
B	HCRIS only	$t = 2$	Vetter Senior Living (100%)
C	HCRIS only	$t = 1$	Abri Health Services (50%); The Senior Care Liquidation Co., Alan Halperin Trustee (50%)
C	HCRIS only	$t = 2$	Abri Care (100%)
D	HCRIS only	$t = 1$	Concordia Lutheran Ministries of Pittsburgh (100%)
D	HCRIS only	$t = 2$	Concordia Care Network (100%)
<i>Panel B: NHC-identified ownership changes not reported as ownership changes in HCRIS</i>			
E	NHC only	$t = 1$	Destefane Richard (100%)
E	NHC only	$t = 2$	Richard J. Destefane Revocable Living Trust (100%)
F	NHC only	$t = 1$	D'Angelo Cara (25%); Durham Deborah (25%); Horace D'Angelo Jr. Operating Entities Dated 01/01/2012 (25%); Tackett Kimberly (25%)
F	NHC only	$t = 2$	Benoit (9.1%); Calumet South (9.1%); Drake Louis Enterprise (9.1%); Fairhome Tr. UAD 12312012 (9.1%); Fairhome UAD 12312012 (9.1%); GZLT (9.1%); Hartman David (9.1%); Hartman Mark (9.1%); Krupp Ari (9.1%); Senderowicz Yossi (9.1%); Willow Delta (9.1%)
G	NHC only	$t = 1$	CMC II Investors (33.3%); Conte Joseph (33.3%); FC LV Holdco (33.3%)
G	NHC only	$t = 2$	Qazi Mohammad (100%)
H	NHC only	$t = 1$	Berkshire Healthcare Systems (67%); Integritus Healthcare (25%); Berkshire Healthcare System (3%)
H	NHC only	$t = 2$	Integritus Healthcare Management Services (100%)
I	NHC only	$t = 1$	DES C 2009 GRAT (100%)
I	NHC only	$t = 2$	DES 2009 Family (100%)

Notes: This table presents examples of discordant ownership-change classifications across HCRIS and the NHC Archive in a panel-style format. Panel A reports cases in which HCRIS reports an ownership change, but the NHC ownership records do not show a clear transfer to a new unrelated controlling owner. Panel B reports cases in which NHC ownership records show large changes in listed owners or ownership shares, but no corresponding HCRIS ownership-change flag is observed. Owner names and shares are taken from the NHC Archive. The labels $t = 1$ and $t = 2$ refer to the ownership records immediately before and after the relevant ownership-record change.

Table 7: Joint Wald Tests of Pre-trends for Monthly Staffing

Outcomes				
	RN	LPN	CNA	Total
Panel A: HPRD				
(1)	192.11 (23) [0.0000]	40.53 (23) [0.0134]	48.47 (23) [0.0015]	53.33 (23) [0.0003]
(2)	46.54 (20) [0.0007]	41.05 (20) [0.0037]	19.06 (20) [0.5178]	28.94 (20) [0.0890]
(3)	19.86 (8) [0.0109]	6.17 (8) [0.6286]	3.53 (8) [0.8968]	6.69 (8) [0.5703]
Panel B: Log(HPRD)				
(1)	45.10 (23) [0.0039]	47.39 (23) [0.0020]	53.22 (23) [0.0003]	44.27 (23) [0.0049]
(2)	29.29 (20) [0.0823]	47.23 (20) [0.0005]	19.58 (20) [0.4845]	26.24 (20) [0.1582]
(3)	4.83 (8) [0.7761]	10.15 (8) [0.2545]	4.73 (8) [0.7865]	7.55 (8) [0.4790]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Specifications: (1) two-year event window, full pre-period; (2) two-year event window excluding the three months preceding transfer; (3) one-year event window excluding the three months preceding transfer.

All specifications include facility and calendar-month fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar month.

Table 8: Joint Wald Tests of Pre-Trends for Quarterly Quality Measures

Outcome	(1)	(2)	(3)
Catheter use	9.33 (7) [0.2299]	9.32 (7) [0.2307]	4.69 (3) [0.1957]
Anti-psychotic medication use	5.07 (7) [0.6510]	5.29 (7) [0.6241]	0.08 (3) [0.9943]
Anti-anxiety/hypnotic medication use	26.33 (7) [0.0004]	28.01 (7) [0.0002]	2.42 (3) [0.4902]
Pressure injuries	2.47 (7) [0.9292]	2.57 (7) [0.9216]	1.48 (3) [0.6872]
Falls with major injury	6.31 (7) [0.5036]	6.96 (7) [0.4327]	2.93 (3) [0.4023]
Weight loss	5.12 (7) [0.6452]	5.24 (7) [0.6312]	2.00 (3) [0.5727]
Decline in physical functioning	10.38 (7) [0.1679]	10.74 (7) [0.1501]	1.90 (3) [0.5930]
Urinary tract infections	13.54 (7) [0.0600]	13.30 (7) [0.0651]	1.29 (3) [0.7322]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Specifications: (1) two-year event window, full pre-period; (2) two-year event window excluding the ownership-change quarter; (3) one-year event window excluding the ownership-change quarter. The ownership-change quarter is excluded because it may combine pre- and post-transfer care, assessment, and documentation. In all three, $\tau = -1$ is the omitted reference period.

All specifications include facility and calendar-quarter fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar quarter.

Pressure injuries is estimated on 2018–2023 only. Other measures use the full sample window.

Table 9: Tests of Differences Between Chain-Affiliated and Independent Facilities

Outcome	Chain – Non-chain
Panel A: Staffing	
RN HPRD	–0.0009 (0.0118)
LPN HPRD	0.0269* (0.0137)
CNA HPRD	0.0440** (0.0216)
Total HPRD	0.0701** (0.0310)
Panel B: Business model	
Occupancy rate	–1.9112*** (0.7221)
Medicare share	0.2802 (0.4691)
Medicaid share	3.7308*** (1.1987)
Average length of stay	14.4865 (10.6701)
Panel C: Quality	
Catheter use	0.2635** (0.0988)
Anti-psychotic medication use	0.5141 (0.4613)
Anti-anxiety/hypnotic medication use	–0.4393 (0.4268)
Pressure injuries	0.2913 (0.2987)
Falls with major injury	0.1258 (0.1417)
Weight loss	–0.5766** (0.2337)
Decline in physical functioning	0.4651 (0.4638)
Urinary tract infections	0.1848 (0.1470)
New antipsychotic medication	–0.0876 (0.1143)
Improved function	–0.4250 (1.1489)

Notes: Each cell reports the coefficient on the interaction between *post* and baseline chain affiliation, with standard errors in parentheses. This coefficient is the difference between the chain and non-chain estimates reported elsewhere in the paper; a negative value indicates a smaller effect among chain-affiliated facilities.

Every right-hand-side term and the calendar-period fixed effects are interacted with chain affiliation, so the specification reproduces the separate subsample regressions exactly while yielding a standard error for their difference. All specifications include facility fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period.

Panels A and B are estimated on the facility-month panel excluding the anticipation window ($\tau = -3, -2, -1$); Panel C on the facility-quarter panel excluding the transition quarter ($\tau = 0$). Facilities with no available chain classification are excluded.

Table 10: TWFE DiD Estimates of *post* on Staffing Outcomes (Stacked Sample, Baseline Donut)

	Outcomes			
	RN	LPN	CNA	Total
HPRD	−0.024*** (0.004)	−0.009 (0.005)	−0.051*** (0.009)	−0.084*** (0.012)
Log(HPRD)	−0.070*** (0.014)	−0.005 (0.008)	−0.027*** (0.004)	−0.027*** (0.004)

Notes: Each cell reports the coefficient on *post* with standard errors in parentheses. The stacked sample includes never-treated and not-yet-treated controls for each cohort; the donut excludes $\tau \in \{-3, -2, -1\}$. Sample sizes: $N_{\text{HPRD}} = 24, 197, 338$; $N_{\text{Log}} = 23, 819, 333$.

Specifications include facility-by-cohort fixed effects (*stack_id*), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Standard errors are clustered two ways by facility and calendar month.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: TWFE estimates of *post* on staffing levels (HPRD): sample-restriction and case-mix robustness.

	N	Outcomes			
		RN	LPN	CNA	Total
(1) Baseline (no anticipation)	550,334	−0.0422*** (0.0058)	0.0021 (0.0063)	−0.0641*** (0.0099)	−0.1041*** (0.0140)
(2) Sample excludes <i>gap</i> > 6	550,175	−0.0422*** (0.0058)	0.0018 (0.0063)	−0.0641*** (0.0099)	−0.1045*** (0.0140)
(3) Sample excludes <i>gap</i> > 3	549,918	−0.0422*** (0.0058)	0.0018 (0.0063)	−0.0641*** (0.0099)	−0.1044*** (0.0139)
(4) Sample excludes <i>gap</i> > 1	546,319	−0.0424*** (0.0058)	0.0022 (0.0063)	−0.0641*** (0.0099)	−0.1042*** (0.0139)
(5) Sample excludes <i>gap</i> > 0	546,319	−0.0424*** (0.0058)	0.0022 (0.0063)	−0.0641*** (0.0099)	−0.1042*** (0.0139)
(6) Drop $t \in \{-4, -3, -2, -1\}$	549,052	−0.0418*** (0.0059)	0.0019 (0.0064)	−0.0640*** (0.0100)	−0.1038*** (0.0141)
(7) Drop $t \in \{-2, -1\}$ only	551,637	−0.0423*** (0.0057)	0.0022 (0.0063)	−0.0637*** (0.0098)	−0.1038*** (0.0138)
(8) + Case-mix: State Quartile	550,334	−0.0444*** (0.0058)	−0.0002 (0.0064)	−0.0683*** (0.0099)	−0.1128*** (0.0139)
(9) + Case-mix: State Decile	550,334	−0.0448*** (0.0058)	−0.0004 (0.0064)	−0.0689*** (0.0099)	−0.1142*** (0.0139)
(10) + Case-mix: National Quartile	550,334	−0.0447*** (0.0058)	−0.0002 (0.0064)	−0.0690*** (0.0099)	−0.1139*** (0.0139)
(11) + Case-mix: National Decile	550,334	−0.0455*** (0.0058)	−0.0005 (0.0064)	−0.0697*** (0.0099)	−0.1158*** (0.0139)

Notes: Each cell reports the *post* coefficient on staffing levels (HPRD), with two-way clustered standard errors (facility and calendar month) in parentheses.

Rows (1)–(7) hold the control set fixed at Spec A (post + number of certified beds, plus facility and calendar-month fixed effects) and vary the sample restriction or anticipation-window definition, as described in each row label. Row (1) is the project’s standard sample: the anticipation window ($\tau = -3, -2, -1$) excluded, no additional gap restriction.

Rows (8)–(11) hold the sample fixed at row (1) and add one case-mix control set on top of Spec A per row – Spec A itself includes no case-mix control, so these rows are not directly comparable to rows (1)–(7). “State Quartile” is the default case-mix control used elsewhere in the paper. Quartile bins use dummies for bins 2–4; decile bins use dummies for bins 2–10 (bin 1 omitted as reference in both cases).

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12: Joint Wald Tests of Pre-Trends: Stacked Event Study (Levels)

	Outcomes			
	RN	LPN	CNA	Total
2 Year Window with Donut	53.28 (20) [0.0001]	32.70 (20) [0.0363]	16.81 (20) [0.6652]	24.73 (20) [0.2121]
1 Year Window with Donut	16.24 (8) [0.0391]	9.44 (8) [0.3068]	8.13 (8) [0.4213]	10.87 (8) [0.2093]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

Sample sizes (stacked rows): 2 Year Window with Donut ($N = 21,332,744$); 1 Year Window with Donut ($N = 11,699,147$).

Stacked design: each treated cohort is compared only to never-treated and not-yet-treated facilities within its own event window; the donut excludes $\tau = -3, -2, -1$ (physically dropped from the estimation sample for treated units).

All specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar month.

8 Robustness

This section reports robustness checks for our staffing results, which are the main input-side estimates in the paper. These checks assess the validity of our identifying assumptions and the sensitivity of our results to alternative modeling and sample choices. Table 11 collects the sample-restriction and control-set exercises; row 1 reproduces the baseline estimates for comparison. Across all exercises the estimated effects retain their sign, statistical significance, and approximate magnitude.

8.0.1 Alternative Comparison Groups

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts (Goodman-Bacon, 2021). To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g . For a given cohort, treated facilities are those with an ownership change in month g , and the control group consists only of facilities that are never treated or not-yet-treated relative to g . We restrict each cohort-specific sample to a symmetric event window around g , apply the same donut exclusion used in the baseline, and then stack these cohort datasets into a single panel. Estimation includes facility-by-cohort fixed effects, calendar-month fixed effects, and cohort fixed effects, with standard errors clustered two ways by facility and calendar month.

Appendix Figures 5a through 5d show that the stacked event-study estimates closely follow the baseline TWFE results: the post-transition dynamics and the relative magnitudes across staffing types are qualitatively unchanged. Appendix Table 10 reports the corresponding average post-change estimates, and Appendix Table 12 reports joint Wald tests of pre-treatment coefficients under the stacked design, which are consistent with flat pre-trends in the remaining pre-period. Together these results indicate that the main findings are not driven by treated-as-control comparisons.

8.0.2 The Anticipation Window

As described in Section ??, the recorded ownership-change date is the closing date, and operational adjustment can begin during the months between contract signing and closing. Our baseline specifications therefore exclude $\tau \in \{-3, -2, -1\}$ from the estimation sample.

Because this exclusion is a modeling choice rather than something the data dictates, we assess it in two ways.

First, the pre-trend evidence in Section 4 indicates that the departure from parallel trends is concentrated in the excluded months rather than diffused across the pre-period. Joint Wald tests of the pre-treatment coefficients improve markedly once these months are removed from the estimation sample. A violation localized to the months immediately preceding closing is what the institutional timeline predicts, and is distinct from a general divergence between treated and untreated facilities that would call the design itself into question.

Second, the estimates are stable to the width of the excluded window. Row 6 of Table 11 widens the exclusion to $\tau \in \{-4, -3, -2, -1\}$ and row 7 narrows it to $\tau \in \{-2, -1\}$. Across both, the estimated effects move by less than 0.001 hours per resident day for every staff category and retain their significance levels. The results are therefore not driven by the width of the window.

8.0.3 Reporting Gaps

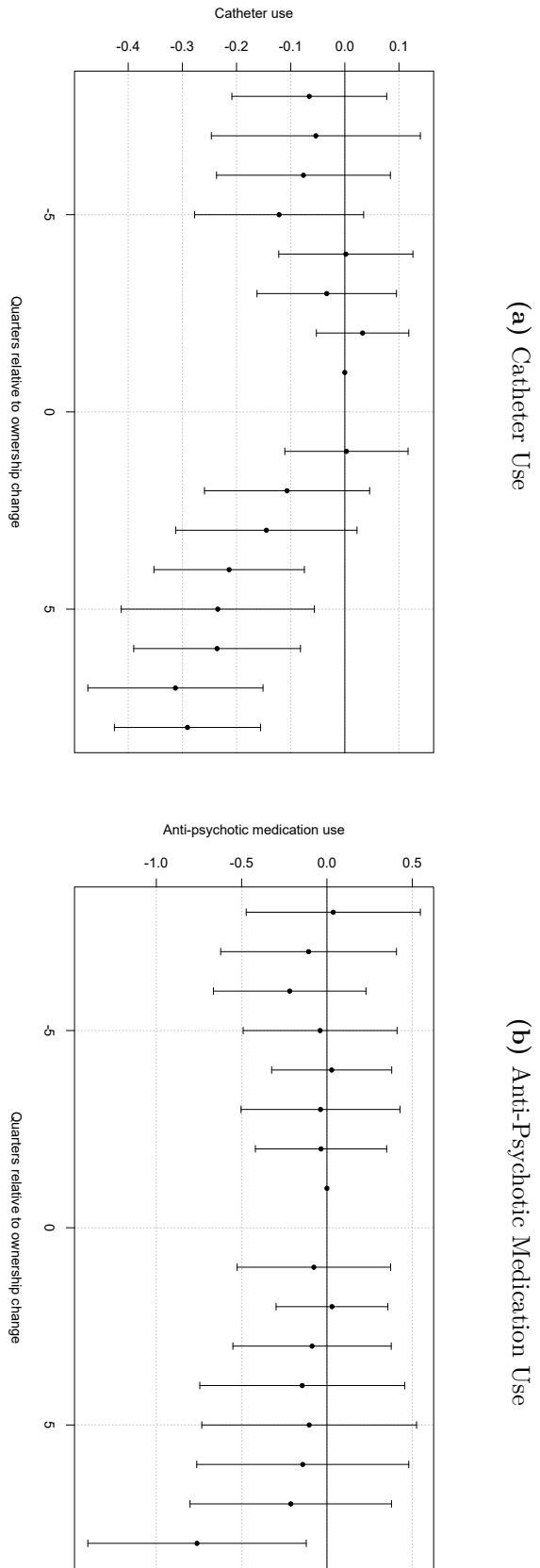
PBJ staffing data are occasionally missing or reported with gaps between consecutive observations. If facilities with intermittent reporting differ systematically in their staffing responses, our estimates could reflect reporting behavior rather than staffing behavior. Rows 2 through 5 of Table 11 progressively restrict the sample to facilities with more continuous reporting, excluding facility-months preceded by a gap exceeding six, three, and one months, and finally excluding any facility-month preceded by a gap at all.

The estimates are unchanged across all four restrictions. RN staffing declines by 0.0368 hours per resident day in the baseline and between 0.0368 and 0.0370 across the restricted samples; total staffing declines by 0.0940 hours in the baseline and by 0.0940 or 0.0941 in every restriction. This is unsurprising given then the strictest restriction removes fewer than one percent of facility-months. Reporting gaps do not drive the staffing results.

8.0.4 Defining Ownership Change

Our ownership-change measure rests on two thresholds described in Section 4: we require more than fifty percent of ownership shares to turn over, and a matching window, under which we require the HCRIS and NHC dates to fall within six months of one another. (I haven't actually done this yet; is it something worth doing?)

Figures



(c) Anti-Anxiety or Hypnotic Medication Use

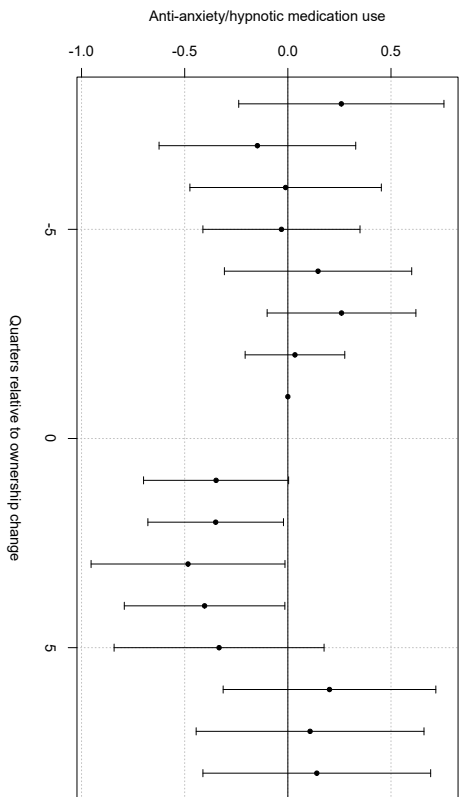
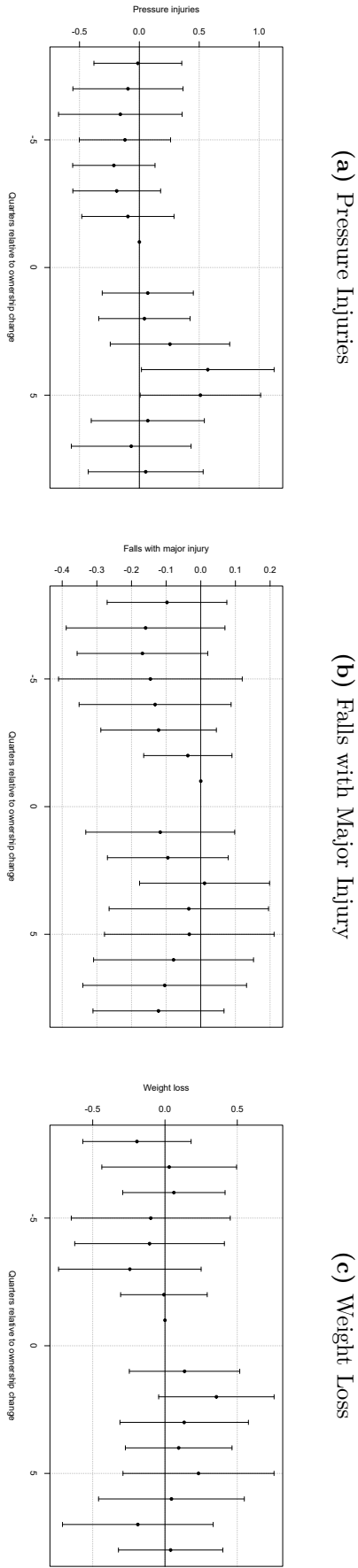


Figure 2: Dynamic Effects of Ownership Change on Labor Saving Mechanisms. Event-study estimates of the effect of a change in ownership on labor saving mechanism measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.



(d) Decline in Physical Functioning

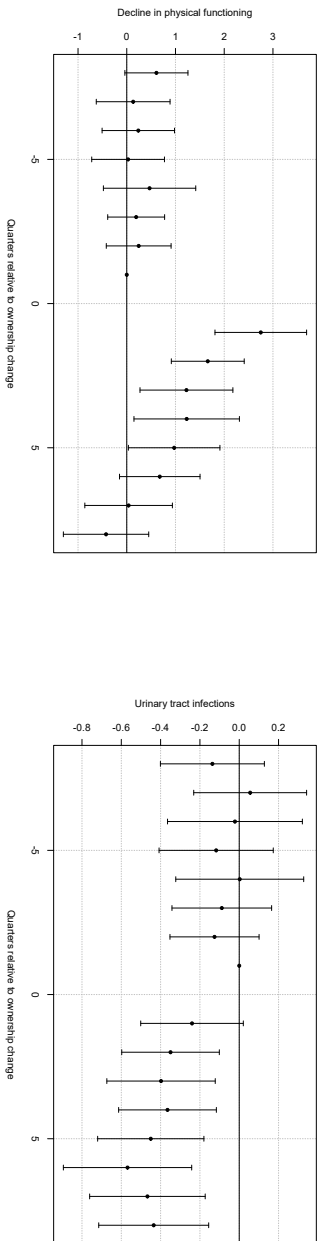


Figure 3: Dynamic Effects of Ownership Change on Resident Outcome Measures. Event-study estimates of the effect of a change in ownership on resident outcome measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.

Geographic Distribution of Ownership Changes

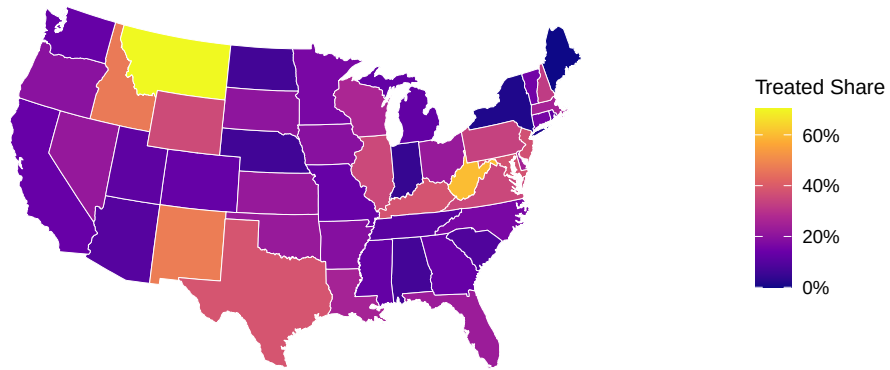
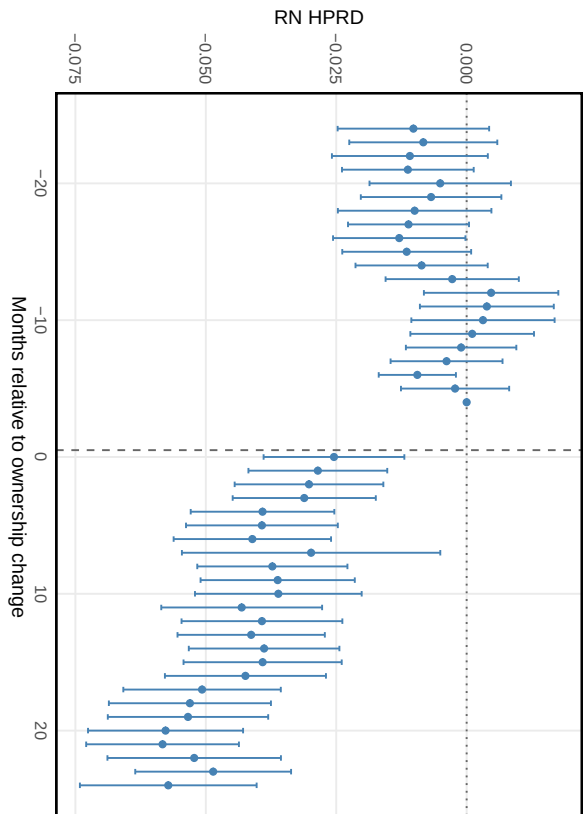
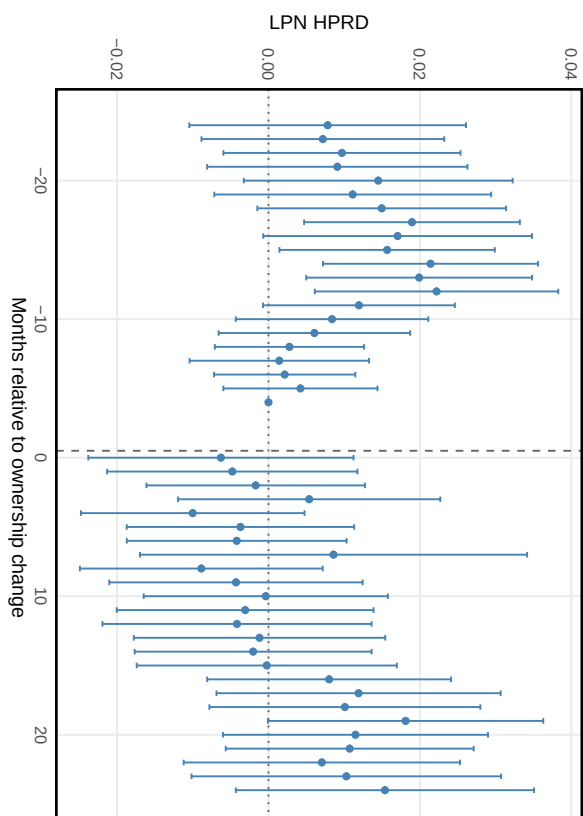


Figure 4: Geographic Distribution of Ownership Changes. The figure reports the share of nursing homes in each state that experienced an ownership change during the study period.

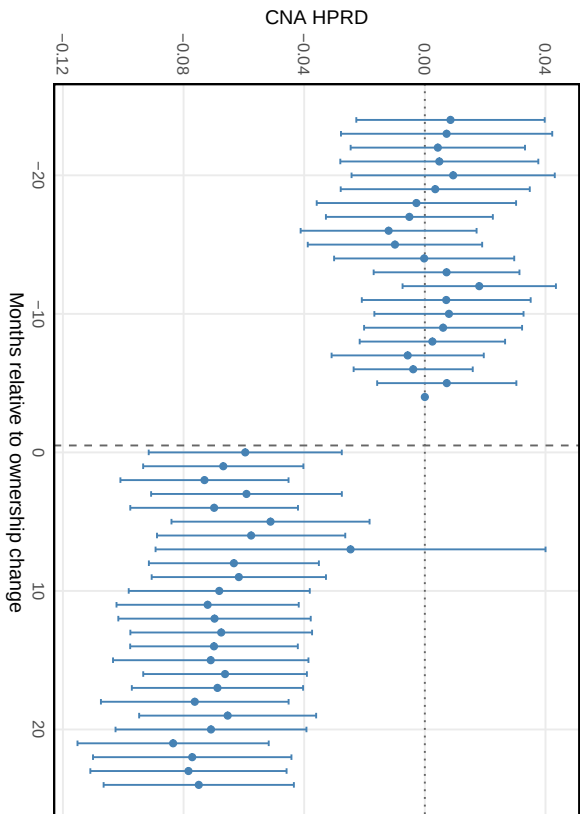
(a) Registered Nurses (RN)



(b) Licensed Practical Nurses (LPN)



(c) Certified Nursing Assistants (CNA)



(d) Total Nursing Hours

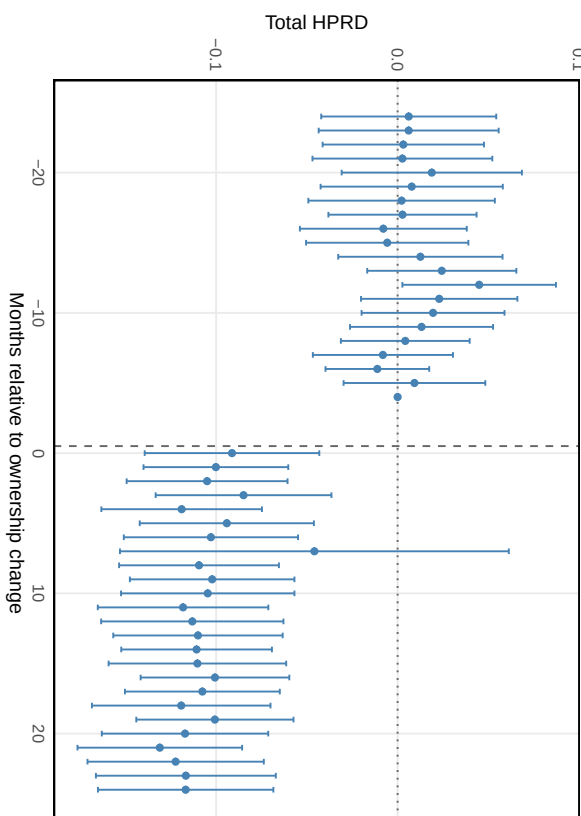


Figure 5: Stacked Event-Study Estimates of Ownership Change on Nursing Staffing. The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; the reference period is $\tau = -4$.